

The result is the waving message animation you see when you run the program.

```
1. """Sine Message, by Al Sweigart al@inventwithpython.com
2. Create a sine-wavy message.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, artistic"""
5.
6. import math, shutil, sys, time
7.
8. # Get the size of the terminal window:
9. WIDTH, HEIGHT = shutil.get_terminal_size()
10. # We can't print to the last column on Windows without it adding a
11. # newline automatically, so reduce the width by one:
12. WIDTH -= 1
13.
14. print('Sine Message, by Al Sweigart al@inventwithpython.com')
15. print('(Press Ctrl-C to quit.)')
16. print()
17. print('What message do you want to display? (Max', WIDTH // 2, 'chars.)')
18. while True:
19.     message = input('> ')
20.     if 1 <= len(message) <= (WIDTH // 2):
21.         break
22.     print('Message must be 1 to', WIDTH // 2, 'characters long.')
23.
24.
25. step = 0.0 # The "step" determines how far into the sine wave we are.
26. # Sine goes from -1.0 to 1.0, so we need to change it by a multiplier:
27. multiplier = (WIDTH - len(message)) / 2
28. try:
29.     while True: # Main program loop.
30.         sinOfStep = math.sin(step)
31.         padding = ' ' * int((sinOfStep + 1) * multiplier)
32.         print(padding + message)
33.         time.sleep(0.1)
34.         step += 0.25 # (!) Try changing this to 0.1 or 0.5.
35. except KeyboardInterrupt:
36.     sys.exit() # When Ctrl-C is pressed, end the program.
```

After entering the source code and running it a few times, try making experimental changes to it. The comments marked with (!) have suggestions for small changes you can make.

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens if you change `math.sin(step)` on line 30 to `math.cos(step)`?
2. What happens if you change `math.sin(step)` on line 30 to `math.sin(0)`?